

SOC-5811 Week 9: Potential outcomes

Nick Graetz

University of Minnesota, Department of Sociology

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REVIEW

Research Question: What is the difference in grandchild wealth by grandparent race, controlling for grandparent wealth?

- ▶ Y = grandchild wealth (continuous)
- ▶ X = grandparent wealth (continuous)
- ▶ Z = grandparent race (categorical: Black, White)

A: $y_i = \beta_0 + \boxed{\beta_1}(X)$

B: $y_i = \beta_0 + \boxed{\beta_1}(X) + \beta_2(Z)$

C: $y_i = \beta_0 + \beta_1(X) + \boxed{\beta_2}(Z)$

D: $y_i = \beta_0 + \beta_1(X) + \beta_2(Z) + \boxed{\beta_3}(X \times Z)$

REVIEW

Research Question: What is the difference in the average association between grandparent wealth and grandchild wealth by race?

- ▶ Y = grandchild wealth (continuous)
- ▶ X = grandparent wealth (continuous)
- ▶ Z = grandparent race (categorical: Black, White)

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- ▶ Y = grandchild wealth (continuous)
- ▶ X = grandparent wealth (continuous)
- ▶ Z = grandparent race (categorical: Black, White)

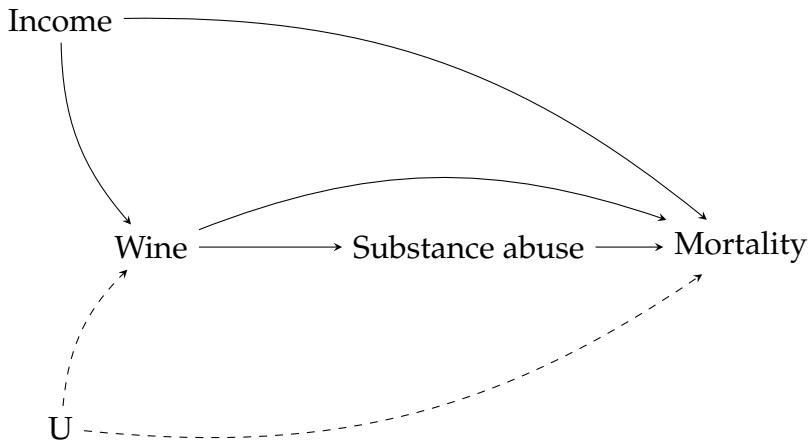
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REVIEW DAGs



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Key steps for building a quantitative research question:

- ▶ **(1)** Draw a conceptual model to guide your research question.



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- ▶ **(3)** Refine conceptual model into a DAG (this may include drawing the “causal field”).



REVIEW DAGs

Key steps for building a quantitative research question:

- ▶ **(1)** Draw a conceptual model to guide your research question.
- ▶ **(2)** Read the literature. What do we know?
- ▶ **(3)** Refine conceptual model into a DAG (this may include drawing the “causal field”).
- ▶ **(4)** Think systematically and explicitly about the potential effect of bias.

DAGs

Let's workshop a DAG on the board for someone's research project.



CAUSAL INFERENCE

- ▶ Causal inference involves counterfactuals.



CAUSAL INFERENCE

- ▶ Causal inference involves counterfactuals.
- ▶ What is a *counterfactual*?



POTENTIAL OUTCOMES

- ▶ Key assumption: “each individual in the population of interest has a potential outcome under each treatment state, even though each individual can be observed in only one treatment state at any given point in time.”



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- ▶ “The fundamental problem of causal inference”



POTENTIAL OUTCOMES

We observe some value of y for every unit i :

$$y_i$$

POTENTIAL OUTCOMES

“The fundamental problem of causal inference”

$$\delta_i = y_i^1 - y_i^0$$



POTENTIAL OUTCOMES

“The fundamental problem of causal inference”

$$\delta_i = y_i^1 - y_i^0$$

- ▶ Experiments do not overcome this problem!
- ▶ The true experiment would be the same individual at the same time.

POTENTIAL OUTCOMES

Group	Y1	Y0
Treatment ($X=1$)	Observable	Counterfactual
Control ($X=0$)	Counterfactual	Observable

POTENTIAL OUTCOMES

Group	Y1	Y0
Treatment (X=1)	$y_i^1 \mid X = 1$	$y_i^0 \mid X = 1$
Control (X=0)	$y_i^1 \mid X = 0$	$y_i^0 \mid X = 0$

POTENTIAL OUTCOMES

Group	Y1	Y0
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- ▶ Counterfactuals exist in theory but are **unobservable**.
- ▶ **Causal inference is a prediction problem!**

AVERAGE TREATMENT EFFECTS

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- ▶ We can't predict individual treatment effects, but we can predict the **average treatment effect (ATE)**.
- ▶ The regression model is one way of generating the **predictions** needed to estimate the ATE.

CAUSAL INFERENCE: PREDICTION PROBLEM

$$\text{prestige}_i = \beta_1 \text{college}_i + \beta_2 \text{age}_i + \beta_3 \text{race}_i + \beta_4 \text{sex}_i + \beta_5 \text{sibs}_i + \beta_6 \text{fpres}_i$$

```
##
## Call:
## lm(formula = prestige10 ~ college + age + as.factor(race) + sex +
##     sibs + papres10, data = gss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -40.745  -8.049  -0.452   7.906  37.196
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   35.48531    1.37482   25.811 < 2e-16 ***
## college       13.49147    0.46057   29.293 < 2e-16 ***
## age           0.08419    0.01315    6.400 1.8e-10 ***
## as.factor(race)2 -1.71212    0.79528   -2.153 0.03141 *
## as.factor(race)3 -0.27320    0.79941   -0.342 0.73257
## sex           -0.68775    0.44159   -1.557 0.11948
## sibs          -0.26907    0.08859   -3.037 0.00241 **
## papres10       0.06071    0.01760    3.449 0.00057 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.82 on 2917 degrees of freedom
## Multiple R-squared:  0.2782, Adjusted R-squared:  0.2765
## F-statistic: 160.6 on 7 and 2917 DF, p-value: < 2.2e-16
```


CAUSAL INFERENCE: PREDICTION PROBLEM

##	college	prestg10	y0	y1
##	<num>	<num>	<num>	<num>
## 1:	0	38	40.33558	53.82706
## 2:	1	57	43.68202	57.17349
## 3:	1	40	35.71267	49.20415
## 4:	0	60	37.58241	51.07388
## 5:	1	60	40.80190	54.29338
## 6:	0	16	37.78251	51.27398

CAUSAL INFERENCE: PREDICTION PROBLEM

```
mean(gss$y1)
```

```
## [1] 54.17606
```

```
mean(gss$y0)
```

```
## [1] 40.68459
```

CAUSAL INFERENCE: PREDICTION PROBLEM

```
mean(gss$y1) - mean(gss$y0)
```

```
## [1] 13.49147
```

CAUSAL INFERENCE: PREDICTION PROBLEM

```
mean(gss$y1) - mean(gss$y0)
```

```
## [1] 13.49147
```

```
summary(mod)
```

```
##
## Call:
## lm(formula = prestg10 ~ college + age + as.factor(race) + sex +
##     sibs + papres10, data = gss)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -40.745  -8.049  -0.452   7.906  37.196
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## Coefficients:
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## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## Residual sum of squares: 11.82 on 2917 degrees of freedom
```

AVERAGE TREATMENT EFFECTS

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$$\beta_1 = \sum_n (y_i|X = 1) - (y_i|X = 0)$$

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- ▶ We can't predict individual treatment effects, but we can predict the **average treatment effect (ATE)**. The regression model is one way of generating the **predictions** needed to estimate the ATE.
- ▶ But when can these **predictions** be interpreted as **potential outcomes**?

AVERAGE TREATMENT EFFECTS

Independence assumption

$$(Y^1, Y^0, \delta) \perp\!\!\!\perp X$$

- ▶ Also referred to as exchangeability, no confounding, etc.

AVERAGE TREATMENT EFFECTS

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$$ATE = E[y_i^1] - E[y_i^0]$$

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Group	Y1	Y0
Treatment (X=1)	Observable	Counterfactual
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$$ATE = E[y_i^1] - E[y_i^0]$$

$$ATT = E[y_i^1 | X = 1] - E[y_i^0 | X = 1]$$

AVERAGE TREATMENT EFFECTS

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$$ATE = E[y_i^1] - E[y_i^0]$$

$$ATT = E[y_i^1 | X = 1] - E[y_i^0 | X = 1]$$

$$\mathbf{ATC} = E[y_i^1 | X = 0] - E[y_i^0 | X = 0]$$

AVERAGE TREATMENT EFFECTS

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$$ATE = E[y_i^1] - E[y_i^0]$$

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$$ATC = E[y_i^1 | X = 0] - E[y_i^0 | X = 0]$$

When will all three of these be identical?

EDUCATION AND EARNINGS

30/30



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Driven to DiscoverSM

